XO-1b

R-0068

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_XO-1_250525` · created 2026-07-15T15:58:45+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460820.8442 +/- 0.0063 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.145 +/- 0.033
Transit depth (Rp/Rs)^2	2.11 +/- 0.97 [%]
Orbital Inclination (inc)	89.55 +/- 1.26
Airmass coefficient 1 (a1)	1.079 +/- 0.026
Airmass coefficient 2 (a2)	-0.059 +/- 0.03
Scatter in the residuals of the lightcurve fit is	2.38 %
Best Comparison Star	#3 - [298, 274]
Optimal Aperture	2.51
Optimal Annulus	14.35
Transit Duration (day)	0.0405 +/- 0.0071

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.145 ± 0.033 vs 0.1326 (deviation 0.22σ)
Duration fit vs published	0.0405 d vs 0.1238 d (deviation 7.02σ)
Mid-time O–C	-15.4 min
Depth SNR	2.6
Residual scatter	2.38 %

Light curve

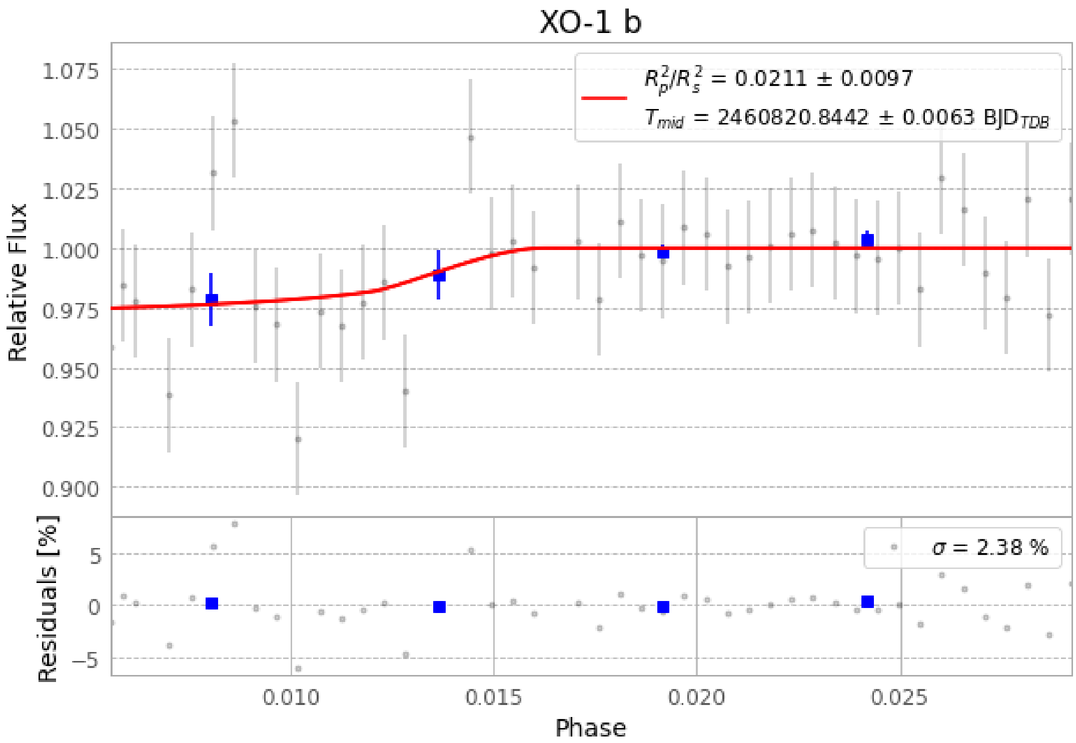

Figure 1 — FinalLightCurve_XO-1 b_2025-05-25.png (SHA-256 109dabb08aa5c138…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [314, 313], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 39bf7799967d8df0…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13`

Data provenance

47 FITS frames (31 MB), `XO-1250525084003.FITS` → `XO-1250525105416.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-XO-1-250525.log (ef9c2f914dad…), FinalLightCurve_XO-1 b_2025-05-25.png (109dabb08aa5…), FinalLightCurve_XO-1 b_2025-05-25.csv (fc376794f2e8…)

Compute resources

Wall time 130.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)